

STUDENT DROPOUT PREDICTION SYSTEM

Introduction

Student dropout remains a significant concern in higher education, as it affects not only institutional performance but also students' future opportunities. Early identification of high-risked dropout students enables timely interventions and improves retention rates. With the growth of Educational Data Mining and Machine Learning approaches such identification is achievable. This Project aims to combine classification, clustering and explainable AI techniques to predict dropout. Using SVM for robust classification of student statuses, BIRCH clustering to group similar profiles, and SHAP values for model interpretability, we not only predict but also explain key drivers of success or failure.

Methodology

The overall workflow of this study follows the CRISP-DM framework, which provides a structured approach to data mining projects. It includes key stages such as data understanding, data preparation, modeling, and evaluation.

The dataset used in this study is a Kaggle dataset which represents the demographic, academic, and financial attributes of students with target distribution of Graduate: 2209(49.93%), Dropout: 1421(32.12%), Enrolled: 794(17.95%). Projects start with preprocessing where data was thoroughly explored through descriptive statistics and structural analysis. Missing values and duplicates were checked and outliers were treated using IQR capping. Further Exploratory Data Analysis was performed in order to get a deeper understanding of the data. Additionally, several meaningful features were engineered, such as overall grade average, grade change, parental education level, financial stability, economic context, and total approved units, to better capture student performance and background. Later, numerical features were scaled to ensure uniformity across variables.

For modeling, the dataset was split into training and testing sets and multiple classification algorithms, including Logistic Regression, Support Vector Machine, Random Forest, K-Nearest Neighbors, and XGBoost, were implemented and evaluated using standard performance metrics. Among these, SVM demonstrated superior performance and was further optimized using hyperparameter tuning with cross-validation and regularization techniques.

To enhance interpretability, SHAP (SHapley Additive exPlanations) was applied to the optimized model, enabling both global and local understanding of feature contributions. Additionally, Unsupervised learning algorithms like Kmeans (K=2) and Birch(K=2, threshold=0.7) were also implemented and parameters were optimised using elbow method and silhouette score. BIRCH was chosen due to its efficiency in handling large datasets and its ability to be incrementally build a hierarchical clustering structure using a CF (Clustering Feature) tree.

Results & Analysis

Outlier detection was performed using the IQR (Interquartile Range) method. Several features showed the presence of significant outliers, including Application Order (~12%), Debtor (~11%), Tuition Fees (~12%), and Scholarship Holder (~25%). Multiple classification algorithms were implemented to predict student status. The models were evaluated using Accuracy, Precision, Recall, and F1-score, with special emphasis on recall (false negatives are costly) and f1-score(because of class imbalance) to effectively identify dropout students.

Model	Accuracy	Precision	Recall	F1-Score
Logistic Regression	0.72	0.70	0.72	0.70
SVM (Untuned)	0.75	0.75	0.75	0.74
SVM (Tuned)	0.79	0.78	0.79	0.78
Random Forest	0.74	0.73	0.74	0.73
XGBoost	0.73	0.72	0.73	0.72
KNN	0.69	0.68	0.69	0.68

To improve model performance, hyperparameter tuning was applied to the SVM model. The optimal parameters obtained were a {'svm__C': 100, 'svm__degree': 2, 'svm__gamma': 0.001, 'svm__kernel': 'rbf'}. In Clustering, the results from BIRCH were consistent with K-Means, producing similar cluster distributions of high-performing and at-risk students.

- Cluster 0 represents low-risk students, who have high academic performance, more approved subjects, and are mostly graduates or continuing students.
- Cluster 1 represents high-risk students, who have low academic performance, fewer approved subjects, and a higher likelihood of dropping out.

Additionally , SHAP analysis showed that academic performance and financial status are the most important factors influencing student dropout prediction.

Conclusion

This study demonstrates that combining machine learning, clustering, and explainable AI provides an effective approach to understanding and predicting student dropout. The SVM model, optimized through tuning, achieved strong performance, particularly in identifying at-risk students. Clustering further revealed meaningful student groups, while SHAP analysis highlighted academic performance especially second semester outcomes as the most critical factor influencing success. These insights can help institutions take timely, data-driven actions to improve student retention.